Multimodal Skin Lesion Classification: Investigating the Influence of Metadata on Model Decision-Making

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1 Introduction and Background

Skin cancer is one of the most common cancer types in the world, making it an important area of research. Early detection is critical to enhance treatment success and reduce the mortality risk [Watson et al., 2026, Jerant et al., 2000, Dildar et al., 2021]. Diagnosis is often based on image analysis, making it a typical use-case for AI-based models such as convolutional neural networks (CNNs) to improve accuracy and efficiency [Musthafa et al., 2024]. A more recent approach is the use of multimodal data, which provides a more detailed view of samples [Hao et al., 2025]. In oncology, multimodal approaches enhance tumor characterization and enable more personalised treatment planning [Hao et al., 2025].
In this work, we investigate how patient metadata influences model decision-making in skin lesion classification by comparing image-only, metadata-only, and multimodal models. This setup allows us to assess the added value of incorporating complementary non-image information. However, performance metrics alone are not sufficient to explain the underlying decision-making mechanisms of the model. Therefore, in addition to analyzing the overall classification results, we investigate which metadata features contribute most strongly to the predictions and how the integration of metadata affects the model's visual attention. We complement the quantitative performance analysis with explainability methods to better understand how the models reached their predictions.

2 Method

2.1 Model Architectures

2.1.1 Image-Only Baseline

To establish a strong image-only baseline, we compared three widely used CNN architectures: VGGNet, GoogLeNet, and ResNet. These architectures were commonly used approaches for skin lesion classification in literature [Romero Lopez et al., 2017, Yu et al., 2025, Wu et al., 2022]. Based on this comparison, we selected the best-performing architecture as the final image-only baseline for the remainder of the study.

Further, we performed minor hyperparameter optimization, including adjustments to the learning rate and batch size to improve model performance.

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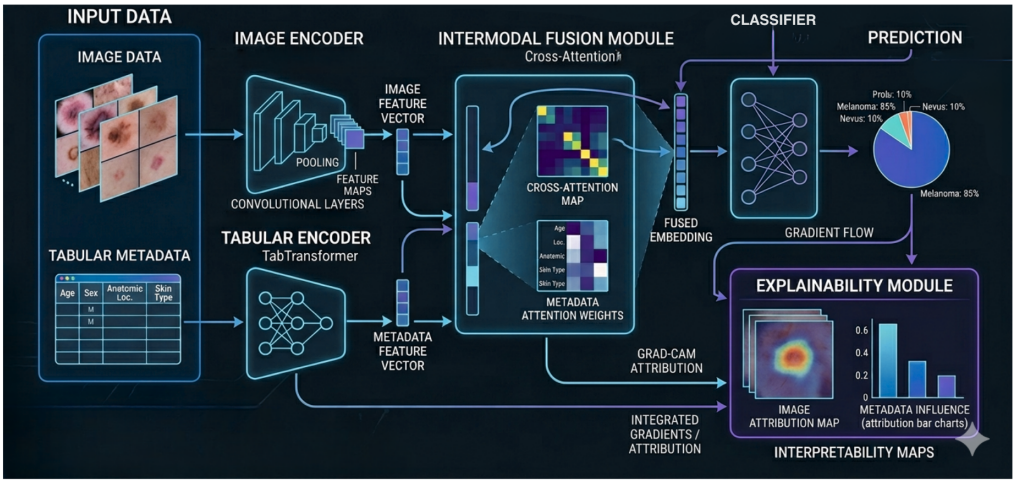

Figure 1: Conceptual overview of the proposed multimodal architecture. Skin lesion images are processed by a CNN-based image encoder, while patient metadata is encoded using a TabTransformer. Both representations are fused through a cross-attention module and passed to a classifier for final prediction. The architecture also facilitates interpretability by enabling analysis of visual attributions and metadata-related attention patterns. The figure was created with AI assistance and manually reviewed and adapted by the authors for technical accuracy.

2.1.2 Metadata-Only Baseline

For the metadata-only baseline, we used a TabTransformer as the main architecture [Huang et al., 2020]. This model was chosen because it represents each metadata feature as an individual token and learns relationships between features through self-attention, rather than processing the metadata as an unstructured flat input. This makes it possible to capture dependencies between variables, which may jointly influence the prediction.

In addition to serving as a strong standalone baseline, this architecture also provides a suitable metadata encoder for the multimodal model. Since the TabTransformer produces contextualised feature representations, it allows the metadata to be integrated more effectively with image features during multimodal fusion.

2.1.3 Multimodal Model

The multimodal model combines the image encoder and the metadata encoder into a joint classification architecture. For the image branch, we used the best-performing CNN architecture selected in the image-only baseline experiments, which extracts visual feature representations from the input images. A conceptual overview of the proposed multimodal architecture is shown in Figure 1.

Both encoded representations from the two encoders are fused using a cross-attention mechanism [Cai et al., 2022]. This allows the image features to attend to the contextualised metadata embeddings and enables the model to determine which metadata variables are most relevant for a given lesion image [Baltrusaitis et al., 2019]. The resulting fused representation is then passed to a classifier that produces the final prediction.

This multimodal design allows the model to combine complementary visual and patient-level information in a flexible way. In addition, the attention weights learned during fusion can be used as a useful signal for analyzing the influence of individual metadata features on the model's decision-making process.

2.2 Evaluation

Our evaluation was designed to address two main questions: whether the inclusion of patient metadata improves predictive performance, and how this additional information influences the model's decision-making process.

To assess predictive performance, we compared the image-only baseline, metadata-only baseline, and multimodal model on the binary classification task using standard metrics, including accuracy, precision, recall, F1-score, and area under receiver operating characteristic curve (AUROC). This comparison allowed us to determine whether combining image and metadata information provides an advantage over unimodal approaches.

To investigate how metadata influences the decision process, we complemented the quantitative evaluation with explainability analyses. First, we examined the cross-attention weights of the multimodal model to identify which metadata features were emphasised for individual predictions. Second, we compared Grad-CAM visualisations of the image-only and multimodal models to analyze whether the inclusion of metadata affected the model's visual attention [Selvaraju et al., 2019]. Together, these analyses allowed us to study not only whether additional information improved classification performance, but also how it altered the model's internal decision-making behaviour.

2.3 Training Setup

Dataset For our analysis, we used the Melanoma Research Alliance Multimodal Image Dataset (MRA-MIDAS) [mid, 2023]. It represents the first publicly available dataset with prospectively collected and systematically paired dermoscopic and clinical images covering multiple skin lesion types [mid, 2023]. The dataset includes 3216 images and metadata from 733 patients and is relatively balanced between benign (1500) and malignant (1419) cases.

The metadata contains patient-level information such as age, ethnicity, and sex, as well as lesion-specific attributes including size, anatomical location, and textual descriptions, and up to three additional clinical impressions per sample. Ground-truth labels are provided, which were used for binary classification into malignant and benign. The labeled data was split into training, validation, and test sets (80%-10%-10%).

Additionally, 497 unlabeled samples were used as an external test set, where a clinical impression served as a proxy label.

2.4 Implementation Details

All models were initialised with ImageNet-pretrained weights for the image encoder. Images were resized and augmented using standard techniques such as random flipping, rotation, and color jitter to improve generalization.

For the metadata, categorical features were embedded and numerical features were normalised before being passed to the TabTransformer. During training, dropout was applied to reduce overfitting.

We used the Adam optimizer with a fixed learning rate and trained the models for a predefined number of epochs, selecting the best model based on validation performance.

3 Results

4 Conclusion

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A Appendix (Optional)